Retain and Enhance Modality-Specific Information for Multimodal Remote Sensing Image Land Use/Land Cover Classification

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***Abstract*—Multimodal remote sensing (RS) image land use/land cover (LULC) classification using optical and synthetic aperture radar (SAR) images has raised attention in recent studies. Current methods primarily employ multimodal fusion operations to directly explore relationships between multimodal features and obtain fused features, leading to the loss of beneficial modality-specific information problem. To solve this problem, this study introduces a multimodal feature decomposition and fusion (MDF) approach combined with a visual state space (VSS) block, namely MDF-VSS block. The MDF-VSS block emphasizes beneficial modality-specific information and perceives shared land cover information through modality-difference and modality-share features, which are then adaptively integrated to obtain discriminative fused features. Based on the MDF-VSS block, an MDF decoder is designed to retain beneficial multi-scale modality-specific information. Then, a multimodal specific information enhancement (MSIE) decoder is designed to perform modality-difference feature-guided auxiliary classification tasks, further enhancing modality-specific information that is expert in classification. Combining the MDF and MSIE decoders, a novel retain–enhance fusion network (REF-Net) is proposed to retain and enhance modality-specific information that benefits classification, thus improving the performance of multimodal RS image LULC classification. Extensive experimental results obtained on three public datasets demonstrate the effectiveness of the proposed REF-Net. The source code will be available at https://github.com/TINYWAI/REF_Net**

***Index Terms*—Land use/land cover (LULC) classification, modality-specific information, multimodal, optical, remote sensing (RS), synthetic aperture radar (SAR).**

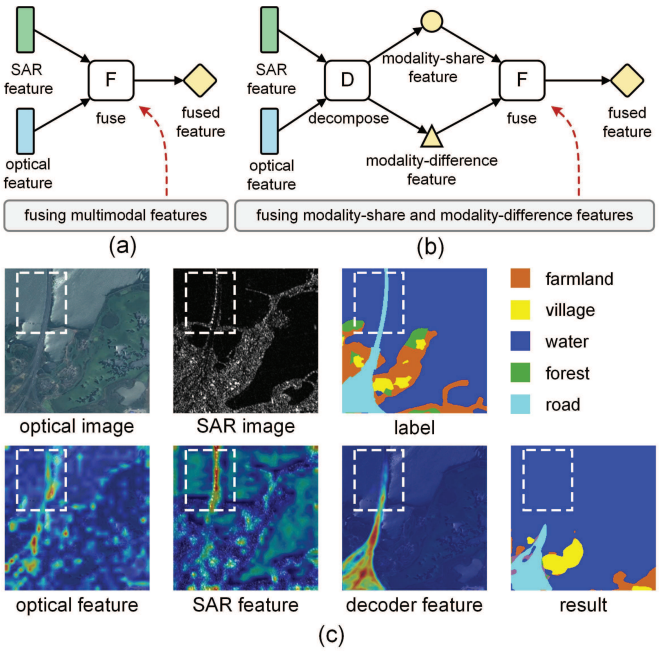


Fig. 1. Conceptual diagrams of (a) deep feature-level fusion method in current methods and (b) designed MDF-VSS block. (c) CAM visualization of challenging area "road" for representative method ASD.

I. Introduction

REMOTE sensing (RS) image land use/land cover (LULC) classification, a fundamental task in Earth observation (EO), involves assigning each pixel in an RS image to a predefined LULC category [1], [2]. This technique is widely applied in urban planning [3], environmental protection [4], and precision agriculture [5]. Optical images often contain rich spectral and texture information but are vulnerable to climatic and lighting conditions, which may lead to difficulties in accurately classifying land objects. In contrast, synthetic aperture radar (SAR) images provide structural and scattering features, enhancing the diversity of representations [6], [7], [8], [9]. Recently, many studies have focused on multimodal RS image LULC classification using both optical and SAR data to overcome performance limitations [10], [11], [12], [13], [14], [15].

Current optical and SAR fusion LULC classification tasks primarily adopt data-driven semantic segmentation methods using deep models [9], [12], [13], [14], [15], which have become the mainstream approach. As illustrated in Fig. 1(a), these methods typically employ multimodal fusion operations, such as concatenation [12], [13], [14] and cross-attention [15], [16], to directly explore relationships between multimodal

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features and obtain fused features. Nonetheless, in challenging scenarios like complex land cover distribution, advanced methods may face performance limitations. Fig. 1(c) shows the classification results of a representative method ASD [12]. As indicated by the white box, while the SAR image captures backscatter information that accurately reflects the road structure, the optical image shows similar spectral properties and smooth textures for both "road" and "water," causing the "road" to be misclassified as "water." Class activation mapping (CAM) visualization shows that the optical feature displays relatively weak responses, whereas the SAR feature exhibits strong and distinct responses for this object. This indicates that SAR-specific information is helpful for classifying the "road" category. However, the responses in the decoder feature are weak and even vanish at the top, resulting in misclassification in the final result. This may suggest a loss of beneficial SAR-specific information. Therefore, current methods have the loss of beneficial modality-specific information problem.

As shown in the white box of Fig. 1(c), the response differences between the optical and SAR features overlap with the region where the decoder exhibits weak responses, suggesting that these differences reflect beneficial modality-specific information. Therefore, measuring the differences between multimodal features helps focus on the beneficial modality-specific information. Motivated by this, this study introduces a multimodal feature decomposition and fusion (MDF) approach combined with a visual state space (VSS) block, namely an MDF-VSS block, to solve the beneficial specific information loss problem. As illustrated in Fig. 1(b), the MDF-VSS block decomposes multimodal features into modality-difference and modality-share features. Here, modality-difference features emphasize modality-specific information that benefits classification. On the other hand, modality-share features capture shared land cover information perceivable by both modalities. The MDF-VSS block then learns fusion weights and adaptively integrates modality-share features into modality-difference features to obtain discriminative fused features. The VSS block unfolds features in different directions to capture contextual information, achieving impressive performance in computer vision tasks [17], [18]. Thus, the MDF-VSS block is built upon VSS blocks [19], to model complex spatial relationships among land cover objects with diverse distribution patterns, thereby enhancing classification performance. Based on the MDF-VSS block, an MDF decoder is designed to control the contribution of beneficial multiscale modality-specific and modality-share information, thereby retaining beneficial modality-specific information from a multiscale perspective.

Then, a multimodal specific information enhancement (MSIE) decoder is designed, consisting of MSIE-OPT and MSIE-SAR decoders, which integrates the decomposed modality-difference features into the optical and SAR features, respectively, to perform optical and SAR auxiliary classification tasks. This selects and emphasizes the specific information that benefits each modality, thereby enhancing the modality-specific that are expert in classification. Combining the MDF and MSIE decoders, a novel retain–enhance fusion network (REF-Net) is proposed to retain and enhance modality-specific information beneficial for classification. This approach overcomes the modality-specific information loss problem and enhances the performance of multimodal RS image LULC classification. In summary, the contributions of this study are summarized as follows.

1) To overcome the loss of beneficial modality-specific information problem, a novel REF-Net is proposed to retain and enhance modality-specific information that is beneficial to classification. REF-Net comprises two key components: the MDF decoder and the MSIE decoder. The MDF decoder adaptively controls the contribution of beneficial multiscale modality-specific and modality-share information, thereby retaining beneficial modality-specific information from a multiscale perspective. The MSIE decoder performs multimodal auxiliary classification tasks to enhance optical- and SAR-specific information that is expert in classification. Therefore, the proposed REF-Net enhances the representation ability of fused features and improves the performance of multimodal RS image LULC classification.
2) To retain modality-specific information that benefits classification, the MDF-VSS block is proposed to emphasize beneficial modality-specific information through modality-difference features while capturing shared land cover information perceivable by both modalities via modality-share features, which are then integrated to obtain discriminative fused features. Specifically, the MDF-VSS block first decomposes multimodal features into modality-difference and modality-share features, which are then adaptively integrated. Building upon the MDF-VSS block, the MDF decoder is proposed to retain beneficial modality-specific information from a multiscale perspective.
3) To enhance modality-specific information that is beneficial to classification, the MSIE decoder comprising MSIE-OPT and MSIE-SAR decoders is proposed to perform modality-difference feature-guided auxiliary classification tasks, thereby enhancing optical- and SAR-specific information that is expert in classification. The MSIE-OPT decoder introduces modality-difference features into optical features to perform optical auxiliary classification tasks, to enhance SAR-specific information. Meanwhile, the MSIE-SAR decoder introduces modality-difference features into SAR features to perform SAR auxiliary classification tasks, to enhance optical-specific information.

Extensive experiments are conducted on three datasets to evaluate the effectiveness of REF-Net. Comparative and visualization experiments demonstrate that REF-Net achieves competitive performance. Feature comparison and analysis intuitively verify that REF-Net effectively decomposes multimodal features and alleviates the loss of beneficial modality-specific information problem.

The remainder of this article is organized as follows. Section II reviews relevant literature on multimodal RS image LULC classification, as well as the application of state space models (SSMs) in visual tasks. Section III provides a comprehensive explanation of the proposed REF-Net. Section IV

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presents the results of comparative experiments conducted on public datasets. Finally, Section V summarizes and concludes the study.

II. Related Work

A. Multimodal RS Image LULC Classification

Regarding multimodal feature extraction, multimodal RS image LULC classification methods can be categorized into model-driven and data-driven approaches.

1) Model-Driven Methods: Many traditional model-driven methods employ morphological operators and subspace learning for multimodal feature extraction and classification [20]. For instance, Rasti et al. [21] utilized extinction profiles (EPs) to extract informative features. Pournemat et al. [22] aligned manifold subspaces in the spectral domain. Hong et al. [23] introduced S2FL, which maps multimodal features into specific and shared spaces to obtain discriminative representations for classification. These methods have been widely applied in multimodal LULC classification using hyperspectral modality [24]. However, these model-driven methods rely on expert knowledge to design operators [25], making it difficult to adaptively handle the complex distributions of land cover in optical-SAR imagery.

2) Data-Driven Methods: Due to the powerful representation ability of deep learning, many researchers have focused on developing data-driven deep learning methods for multimodal RS image LULC classification. These methods can be categorized into two types: patch-based methods and semantic segmentation methods [20].

Patch-based methods typically use a sliding window approach to classify image patches, assigning the classification result to the pixel at the center of each window. Hong et al. [20] first introduced a unified multimodal framework that investigates various fusion strategies for multimodal RS data. Yao et al. [26] developed the transformer-based ExViT, which effectively captures both spatial and channel information. Chang et al. [27] proposed DSymFuser, enabling multilevel local–global feature extraction and cross-modal feature interaction. Although patch-based methods have achieved notable improvements, authoritative studies [28], [29] have pointed out that these methods can be computationally expensive and time-consuming, particularly when applied to optical and SAR data with large size.

Semantic segmentation methods assign a semantic label to each pixel and have become the mainstream approach in optical and SAR fusion for LULC classification [12], [13], [14], [15]. V-FuseNet, introduced by Audebert et al. [29], incorporated a third branch with a residual operation to learn fused features with robust representations. Li et al. [15] proposed MCANet, which employed carefully designed cross-attention and low-high level feature fusion modules to enhance long-distance contextual information, but overlooked the role of spatial representation in feature fusion. Accordingly, JoiTriNet, proposed by Liu et al. [13], utilized channel attention followed by spatial attention on unimodal features to aggregate complementary spectral and structural information. Unlike attention-based methods, ASD proposed by Li et al. [12] introduced a spatially aware circular module and a channel shuffle module to integrate cross-modal spatial and channel representations. Recently, Xu et al. [14] developed SAMFNet, which applied a symmetric attention module with strip pooling to achieve long receptive fields, thereby obtaining complementary and contextual information. Current methods often employ multimodal fusion operations to directly explore the relationships between multimodal features and obtain fused features. However, these methods suffer from a beneficial modality-specific information loss problem.

More recently, several large-scale multimodal RS foundation models have been proposed [30]. Skysense [31] is a billion-scale model pretrained on a curated multimodal RS dataset containing 21.5 million temporal sequences. This method employs multimodal contrastive learning to obtain generalized shared features that can be applied to specific tasks. DOFA [32] enables a single Transformer to adaptively process multimodal data, excelling in 14 EO tasks. This method adopts a unifying and adjusting representation learning to dynamically generate modality-specific features for different tasks. Compared to them, the proposed REF-Net differs in the utilization of shared and specific features. REF-Net emphasizes beneficial modality-specific information and perceives common information through modality-difference and modality-share features. These features are then fused to obtain discriminative fused features, aiming to address the problem of lost beneficial modality-specific information.

B. SSM in Visual Tasks

The SSM concept was initially introduced by the S4 model [33] and subsequently advanced by the S5 model [34] and H3 model [35], aiming to manage long-range contextual information with linear computational cost. Gu and Dao [36] introduced the SSM layer and developed a language model backbone named Mamba, which outperformed Transformers [37] on large-scale natural language data. Recently, Liu et al. [19] extended the SSM block to visual tasks by introducing the 2-D selective scan (SS2D) module, resulting in the VSS block and VMamba architecture, which yielded superior performance in modeling global spatial context. Inspired by this work, many studies have explored the capability of SSM in complex context modeling in visual tasks, including classification [38], object detection [17], segmentation [39], and image restoration [18], achieving competitive performance.

Unlike natural images, RS images typically contain multiple land objects at varying scales, requiring models to possess the capability for complex context modeling and a global perspective for accurate interpretation. To extract deep representations rich in spatial relationships and enhance interpretation performance, some pioneering studies have introduced SSM in RS image processing. Chen et al. [40] introduced RSMamba based on the VSS block for RS image scene classification, highlighting its advantages in global receptive field and linear modeling complexity. Zhao et al. [41] introduced RS-Mamba for dense prediction. The proposed OSSM captures spatial features of land covers in multiple directions, effectively modeling the global context of RS images. ChangeMamba [42]

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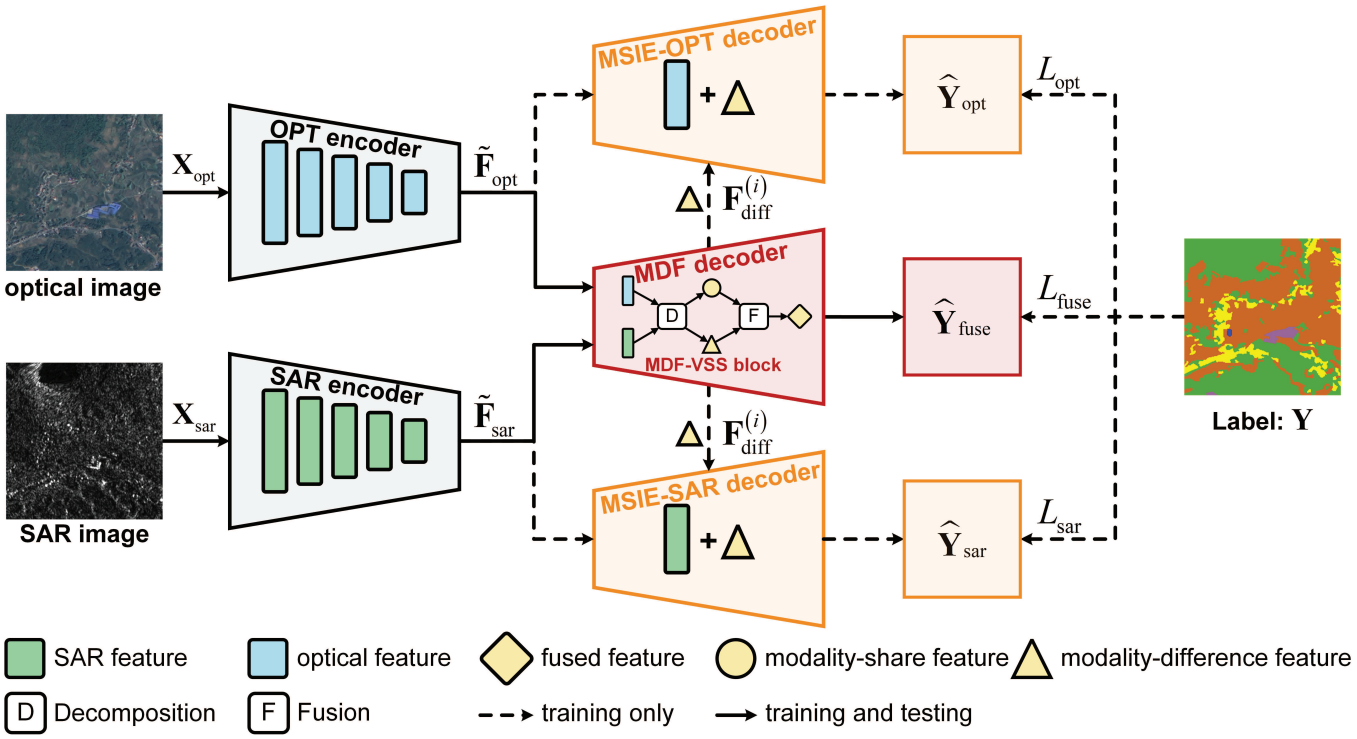


Fig. 2. Pipeline of the proposed REF-Net. Optical and SAR images are first fed into the pseudo-Siamese feature extractors to obtain modality-specific features, i.e., optical and SAR features. These features are then projected to fixed dimensions. Subsequently, the projected multimodal features are fed into the MDF decoder with the MDF-VSS block. The MDF-VSS block applies decomposition and fusion on the multimodal features to obtain modality-difference and modality-share features and then fuses them. The MDF decoder integrates adjacent fused features and ultimately outputs the classification result. The modality-difference features, along with the projected optical and SAR features, are then fed into the MSIE-OPT and MSIE-SAR decoders to perform multimodal auxiliary classification tasks during training, aiming to enhance modality-specific information that is expert in classification.

employed the VSS block to model global spatial context and spatio-temporal relationships across multitemporal RS images for change detection. Ma et al. [43] proposed RS^{3}Mamba for semantic segmentation, incorporating a VSS-based auxiliary branch to provide additional global information. Wang et al. [44] studied PyramidMamba with a designed Mamba-based decoder, which used the selective characteristic of Mamba to reduce homogeneous semantic information. Liu et al. [45] developed CM-UNet, which integrates channel and spatial attention into the Mamba block via residual learning to extract global contextual information. Liu et al. [46] proposed RSCaMa for the RS change captioning task. This work introduced SD-SSM and TT-SSM, both based on SSM, to enhance spatial change perception and temporal understanding.

The successful application of SSM-based methods has demonstrated their ability to model global spatial context. Therefore, this study leverages the VSS block to capture the global spatial context of complex land objects.

III. Proposed REF-Net

The REF-Net is proposed to retain and enhance modality-specific information that is beneficial to classification, which is built upon MDF-VSS blocks and consists of an MDF decoder and an MSIE decoder. The MDF-VSS block emphasizes beneficial modality-specific information through modality-difference features. It also captures shared land cover information perceivable by both modalities via modality-share features. These features are then adaptively integrated to obtain discriminative fused features. Building upon the MDF-VSS block, the MDF decoder is proposed to retain beneficial modality-specific information from a multiscale perspective. Furthermore, the MSIE decoder performs modality-difference feature-guided auxiliary classification tasks, to enhance optical- and SAR-specific information that is expert in classification. Therefore, the proposed REF-Net enhances the representation ability of fused features and improves the performance of multimodal RS image LULC classification.

A. Architecture of REF-Net

Fig. 2 shows the pipeline of the proposed REF-Net, which consists of pseudo-Siamese feature extractors, the MDF decoder, and the MSIE decoder. The MDF decoder, built upon the MDF-VSS block, adaptively controls the contribution of beneficial multiscale modality-specific and modality-share information, ensuring that beneficial multiscale modality-specific information is retained. Meanwhile, the MSIE decoder performs modality-difference feature-guided auxiliary classification tasks during training, to enhance optical- and SAR-specific information that is expert in classification.

For an optical image $\mathbf{X}_{\text{opt}}$ and a SAR image $\mathbf{X}_{\text{sar}}$, the REF-Net pipeline feeds these multimodal data into pseudo-Siamese feature extractors to obtain modality-specific features, i.e., optical features $\mathbf{F}_{\text{opt}} = \{\mathbf{F}_{\text{opt}}^{(i)}\}_{i=1}^{5}$ and SAR features

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$\mathbf{F}_{\text{sar}} = \{\mathbf{F}_{\text{sar}}^{(i)}\}_{i=1}^{5}$, where i denotes the layer index for feature extractors. Then, the optical features $\mathbf{F}_{\text{opt}}$ and SAR features $\mathbf{F}_{\text{sar}}$ are projected into the same dimension, which is empirically set to 128. These projected multimodal features are denoted as $\widetilde{\mathbf{F}}_{\text{opt}} = \{\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}\}_{i=1}^{5}$ and $\widetilde{\mathbf{F}}_{\text{sar}} = \{\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}\}_{i=1}^{5}$.

For each layer i, the projected multimodal features $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ and $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$ are input into the MDF decoder based on MDF-VSS block. The MDF-VSS block decomposes the multimodal features into modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ and modality-share features $\mathbf{F}_{\text{share}}^{(i)}$, and then learns weights for fusing these two parts to obtain fused features $\mathbf{F}_{\text{fuse}}^{(i)}$. The MDF decoder integrates adjacent fused features and outputs the final classification result $\widehat{\mathbf{Y}}_{\text{fuse}}$. The MDF decoder adaptively controls the contribution of beneficial multiscale modality-specific and modality-share information, thereby effectively retaining beneficial modality-specific information from a multiscale perspective.

Then, the decomposed modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ and the projected multimodal features, $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ and $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$, are input into the MSIE decoder. This decoder performs modality-difference feature-guided auxiliary classification tasks and produces results $\widehat{\mathbf{Y}}_{\text{opt}}$ and $\widehat{\mathbf{Y}}_{\text{sar}}$, respectively. The purpose is to enhance SAR- and optical-specific information that is expert in classification.

The REF-Net is optimized in an end-to-end scheme, which simultaneously achieves retaining and enhancing modality-specific information, thus improving the performance of multimodal RS image LULC classification.

B. MDF Decoder

To retain multiscale modality-specific information beneficial for classification, the MDF decoder, based on the MDF-VSS block, is proposed to adaptively control the contribution of beneficial modality-specific and modality-share information from a multiscale perspective. In this decoder, the MDF-VSS block emphasizes beneficial modality-specific information and perceives shared land cover information through modality-difference and modality-share features, which are then fused to obtain fused features. Besides, the VSS block is utilized to capture the global spatial context of complex land objects.

Land cover objects exhibit diverse distribution patterns that provide rich contextual information, which needs to be captured for accurate classification [1]. Recently, the VSS block [19], equipped with an SS2D mechanism, has been introduced for visual data tasks. It unfolds features across different spatial directions, enabling the exploration of inherent relationships within an image while extracting complex global contextual information. The VSS block has shown impressive performance in computer vision tasks [17], [18] and RS image interpretation [40], [41], [42], [43].

Structures of the MDF decoder and MDF-VSS block are shown in Figs. 3 and 4.

1) MDF-VSS Block: To retain modality-specific information that benefits classification, the MDF-VSS block is proposed. It emphasizes beneficial modality-specific information through modality-difference features. It also captures shared land cover information perceivable by both modalities via modality-share features. These features are then integrated to obtain discriminative fused features. Specifically, the MDF-VSS block decomposes multimodal features by measuring their discrepancies to capture modality-difference features, while simultaneously mapping them into a common embedding space to obtain modality-share features. Next, the MDF-VSS block concatenates the modality-share features and adaptively integrates them with the modality-difference features to obtain fused features. As shown in Fig. 4, modality-difference, modality-share, and fused features are represented by a yellow triangle, circle, and rhombus, respectively.

First, the modality-difference features are obtained by applying a subtraction operation to the optical and SAR features, followed by a VSS block. The subtraction operation, a simple and intuitive approach for obtaining feature differences, is motivated by its widespread use in change detection to extract multitemporal feature differences [47], [48]. Then, the VSS block elaborately extracts contextual information related to the differences between modalities, thereby enhancing the representation ability of beneficial difference features, to emphasize modality-specific information that benefits classification. Therefore, modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ are obtained as follows:

$$\mathbf{F}_{\text{diff}}^{(i)} = \text{VSS}\left(\widetilde{\mathbf{F}}_{\text{opt}}^{(i)} - \widetilde{\mathbf{F}}_{\text{sar}}^{(i)}\right). \tag{1}$$

Next, the modality-share features are obtained by mapping multimodal features into a common embedding space [49], [50] using shared-weight VSS blocks, which scan and merge multimodal features pixel by pixel to capture shared land cover information perceivable by both modalities, to enhance the representation ability of shared features. Therefore, for optical and SAR features, $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ and $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$, modality-share features $\mathbf{F}_{\text{share}}^{(i)}$ can be obtained as follows:

$$\mathbf{F}_{\text{share}}^{(i)} \in \left\{\text{VSS}\left(\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}\right), \text{VSS}\left(\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}\right)\right\} \tag{2}$$

where VSS$(\cdot)$ denotes shared-weight VSS blocks.

Finally, the modality-share features $\mathbf{F}_{\text{share}}^{(i)}$ are first concatenated, obtaining aggregated shared features $\mathbf{F}_{\text{share_agg}}^{(i)}$, then integrated into the modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ through a convolutional block structured as convolution-batchnorm-ReLU, to establish the linkage between modality-share features while adaptively learning the fusion weights for shared and difference features. The fused features are obtained by applying a residual block to the integrated features, increasing representation ability. Thus, the fused features $\mathbf{F}_{\text{fuse}}^{(i)}$ can be obtained as follows:

$$\begin{aligned}\mathbf{F}_{\text{share_agg}}^{(i)} &= \text{Concat}\left(\text{VSS}\left(\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}\right), \text{VSS}\left(\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}\right)\right)\\ \mathbf{F}_{\text{fuse}}^{(i)} &= \text{Res}\left(\text{Conv}\left(\mathbf{F}_{\text{share_agg}}^{(i)}\right) + \mathbf{F}_{\text{diff}}^{(i)}\right)\end{aligned} \tag{3}$$

where Concat$(\cdot)$ represents the concatenation operation, Conv$(\cdot)$ denotes the convolutional block comprising a 3×3 convolution, batchnorm, and a ReLU function, and Res$(\cdot)$ represents the residual block.

2) MDF Decoder Based on MDF-VSS Blocks: To retain multiscale modality-specific information beneficial to classification, the MDF decoder, based on the MDF-VSS block, is

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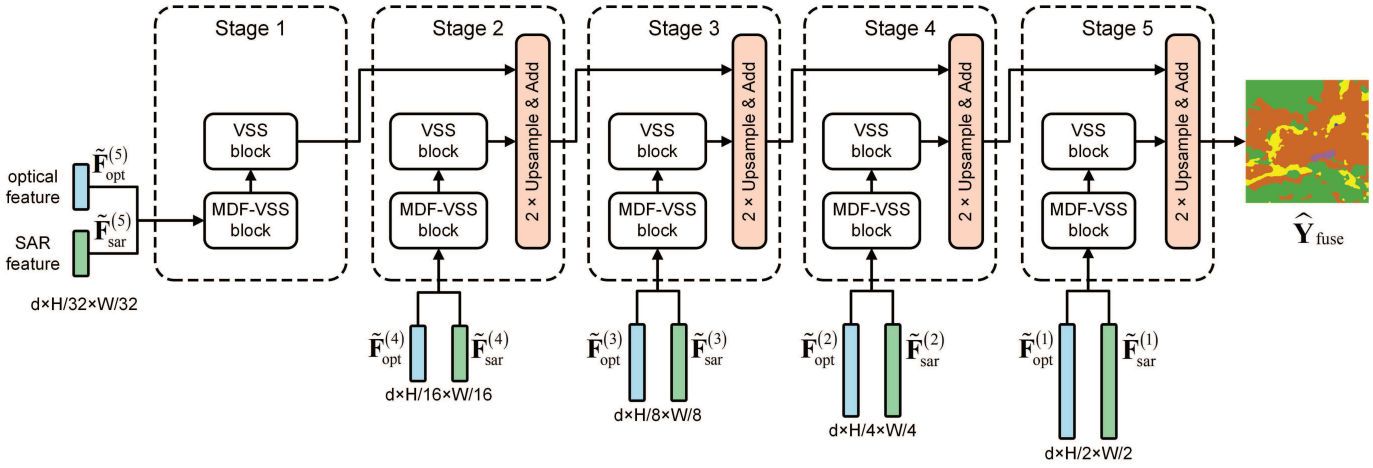


Fig. 3. Specific structure of the MDF decoder based on MDF-VSS block, which involves five stages. For each stage i, optical and SAR features, $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ and $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$, are fed into the MDF-VSS block followed by a VSS block. These features are then upsampled and integrated with adjacent high-resolution features and finally output final classification results $\widehat{\mathbf{Y}}_{\text{fuse}}$.

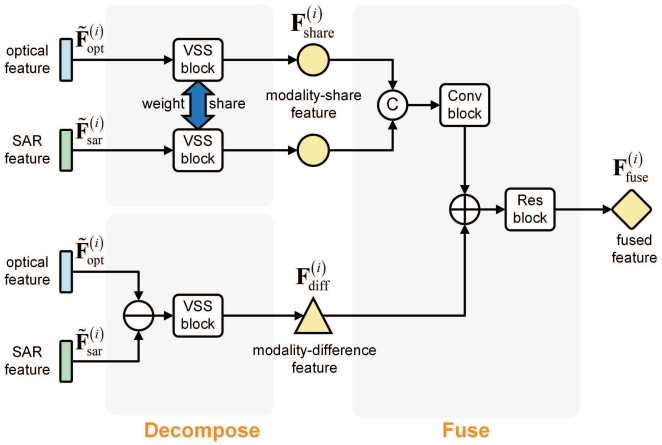


Fig. 4. Specific structure of the MDF-VSS block. Multimodal features $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ and $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$ are subtracted and modeled by a VSS block to obtain modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$. Meanwhile, multimodal features are fed into two shared-weight VSS blocks to obtain modality-share features $\mathbf{F}_{\text{share}}^{(i)}$. Then, modality-share features are concatenated and integrated with modality-difference features using a convolutional block to adaptively learn the fusion weights. Finally, a residual block is utilized to produce fused features $\mathbf{F}_{\text{fuse}}^{(i)}$.

proposed to control the contribution of beneficial multiscale modality-specific and modality-share information.

The structure of the MDF decoder is shown in Fig. 3, which involves five stages to fuse optical and SAR features and performs upsample operations to match the resolution of different stages. For each stage i, optical and SAR features of the same resolutions, $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ and $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$, are fed into the MDF-VSS block to obtain the fused feature $\mathbf{F}_{\text{fuse}}^{(i)}$. Then, a VSS block is adopted to capture the global spatial context of complex land objects. These features are upsampled using bilinear interpolation and integrated with adjacent high-resolution features using a residual block. Finally, the MDF decoder outputs classification results $\widehat{\mathbf{Y}}_{\text{fuse}}$, which are supervised by labels $\mathbf{Y}$ using the cross-entropy loss L_{fuse}, defined as follows:

$$L_{\text{fuse}} = \text{CE}\left(\mathbf{Y}, \widehat{\mathbf{Y}}_{\text{fuse}}\right) \tag{4}$$

where CE denotes the cross-entropy loss.

C. MSIE Decoder

To enhance modality-specific information beneficial to classification, the MSIE decoder is proposed. It performs modality-difference feature-guided auxiliary classification tasks during training. This process enhances optical- and SAR-specific information that is expert in classification. Specifically, the MSIE-OPT decoder introduces modality-difference features into optical features, while the MSIE-SAR decoder introduces modality-difference features into SAR features. The MSIE-OPT and MSIE-SAR decoders perform optical and SAR auxiliary classification tasks, respectively, to enhance SAR- and optical-specific information that is expert in classification, thereby enhancing modality-specific information beneficial to classification.

1) MSIE-OPT Decoder: To enhance SAR-specific information that is expert in classification, the MSIE-OPT decoder is proposed to introduce modality-difference features into optical features through an optical auxiliary classification task.

The structure of the MSIE-OPT decoder is shown in Fig. 5(a), which involves five stages to fuse optical and modality-difference features. For each stage i, modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ are summed to optical features $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ with the same resolutions. Beneficial SAR-specific information is adaptively selected and enhanced through a residual block. Then, features from each stage are upsampled using bilinear interpolation and integrated with adjacent high-resolution features using a residual block. Finally, the MSIE-OPT decoder outputs classification results $\widehat{\mathbf{Y}}_{\text{opt}}$, which are supervised by labels $\mathbf{Y}$ with the cross-entropy loss L_{opt}, defined as follows:

$$L_{\text{opt}} = \text{CE}\left(\mathbf{Y}, \widehat{\mathbf{Y}}_{\text{opt}}\right). \tag{5}$$

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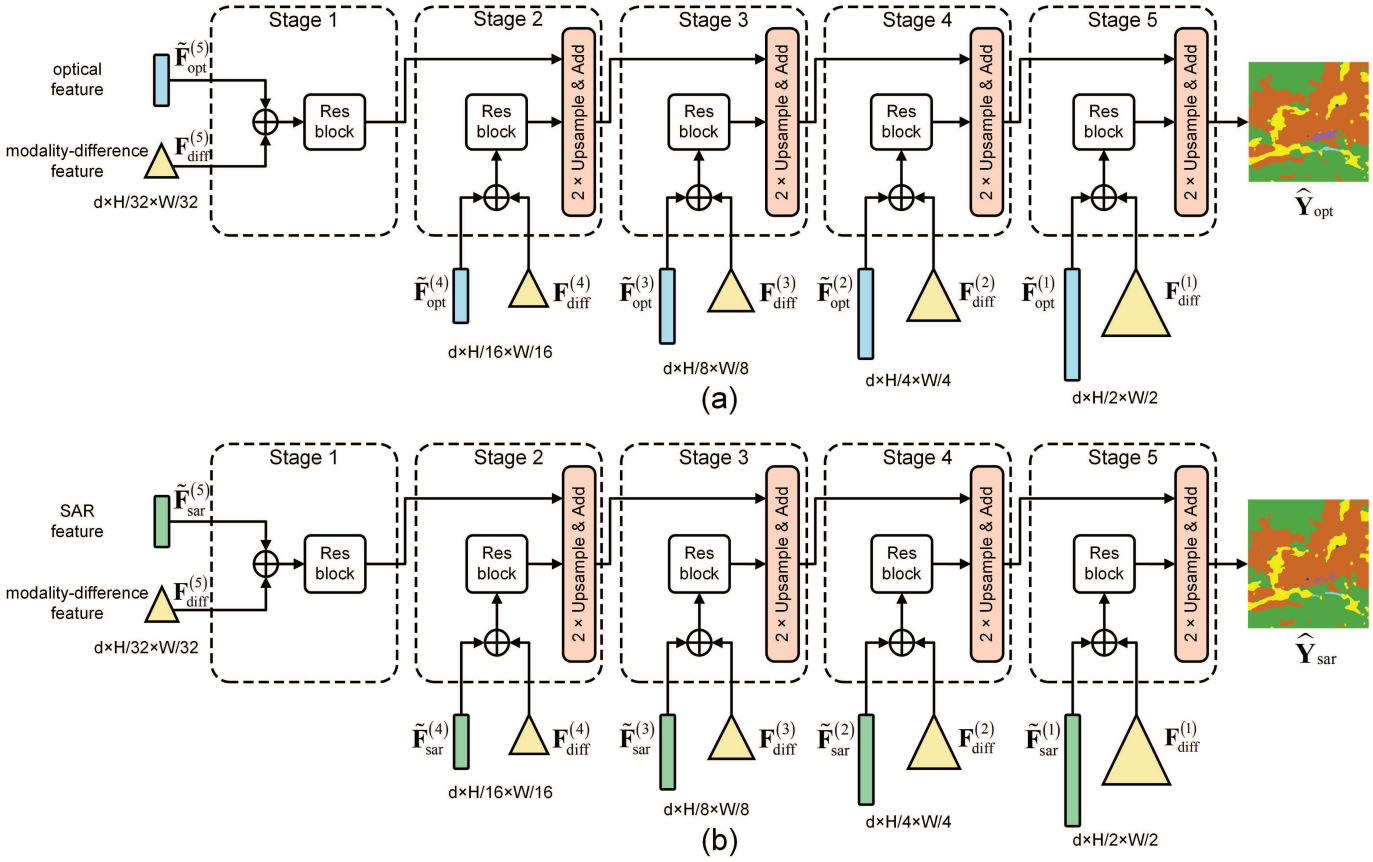


Fig. 5. Specific structure of MSIE decoders, which consists of (a) MSIE-OPT decoder and (b) MSIE-SAR decoder with the same structure. MSIE-OPT and MSIE-SAR decoders both involve five stages. For each stage i, modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ are introduced into optical features $\widetilde{\mathbf{F}}_{\text{opt}}^{(i)}$ or SAR features $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$ followed by a residual block. These features are then upsampled and integrated with adjacent high-resolution features and finally outputs classification results $\widehat{\mathbf{Y}}_{\text{opt}}$ and $\widehat{\mathbf{Y}}_{\text{sar}}$ for multimodal auxiliary classification tasks during training.

2) MSIE-SAR Decoder: To enhance optical-specific information that is expert in classification, the MSIE-SAR decoder is proposed to introduce modality-difference features into SAR features through an SAR auxiliary classification task.

The structure of the MSIE-SAR decoder is shown in Fig. 5(b), which is similar to the MSIE-OPT decoder. In contrast, MSIE-SAR decoder introduces modality-difference features $\mathbf{F}_{\text{diff}}^{(i)}$ into SAR features $\widetilde{\mathbf{F}}_{\text{sar}}^{(i)}$. The MSIE-SAR decoder outputs classification results $\widehat{\mathbf{Y}}_{\text{sar}}$, which are supervised by labels $\mathbf{Y}$ using the cross-entropy loss L_{sar}, defined as follows:

$$L_{\text{sar}} = \text{CE}\left(\mathbf{Y}, \widehat{\mathbf{Y}}_{\text{sar}}\right). \tag{6}$$

D. Loss Function

REF-Net utilized an end-to-end training scheme, simultaneously optimizing MDF and MSIE decoders. As introduced above, the outputs of MDF and MSIE decoders are supervised by three loss functions, which are L_{fuse}, L_{opt}, and L_{sar}. By combining these loss functions, the REF-Net performs an end-to-end training scheme. The formulation of the overall loss is calculated as follows:

$$L = L_{\text{fuse}} + \lambda\left(L_{\text{opt}} + L_{\text{sar}}\right) \tag{7}$$

where λ is a trade-off hyperparameter to balance the contribution of modality-specific information enhancement. According to Section IV-C.4, $\lambda = 0.005$ is the optimal value, which is set for all experiments.

IV. Experiment

A. Dataset Description

The proposed REF-Net is evaluated over three multimodal RS image LULC datasets involving optical and SAR modalities, which are DFC20, WHU-OPT-SAR, and C2Seg-BW datasets.

1) DFC20: The DFC20 dataset [10], curated for the 2020 IEEE GRSS Data Fusion Contest, is a multimodal RS image land cover dataset comprising 6114 pairs of optical and SAR images, each with a resolution of 256 × 256 pixels. The optical images are multispectral data captured by Sentinel-2, covering 13 bands, of which only the RGB bands are used. The SAR images are dual-polarization data from Sentinel-1, including the VV and VH bands. Sentinel images have a native resolution of approximately 10–20 m/pixel, with all data in this dataset resampled to a ground sampling distance of 10 m. The dataset focuses on eight land cover categories: "Forests," "Shrublands," "Grasslands," "Wetlands," "Croplands," "Urban/build_up," "Barren," and "Water." The dataset was split into three subsets with a ratio of 6:2:2,

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resulting in 3668 pairs for training, 1223 pairs for validation, and 1223 pairs for testing.

2) WHU-OPT-SAR: The WHU-OPT-SAR dataset [15] is a multimodal RS land use dataset comprising 100 pairs of optical and SAR images, each with a resolution of 5556 × 3704 pixels, covering an area of approximately 50 000 km^2 in Hubei Province, China. The optical images are multispectral data captured by GF-1, including RGB and near-infrared bands, with a ground resolution of 2 m. The SAR images are acquired by GF-3 Fine Strip II with a sampling resolution of 5 m. The optical images are resampled to a ground resolution of 5 m using bilinear interpolation to achieve pixel-by-pixel registration with the SAR images. The dataset includes seven land use categories: "farmland," "city," "village," "water," "forest," "road," and "others." The dataset was split into three subsets with a ratio of 6:2:2, and the images were cropped into 256 × 256 pixel patches without overlap. As a result, the training, validation, and testing sets contained 17 640, 5880, and 5880 pairs, respectively.

3) C2Seg-BW: The C2Seg-BW dataset [11] is a multimodal RS image LULC classification dataset, comprising images of 5556 × 3704 pixels for Beijing and 6225 × 8670 pixels for Wuhan. The dataset includes hyperspectral, multispectral, and SAR modalities, with this study utilizing multispectral and SAR data for experiments. The SAR data were acquired by GF-3 with a spatial resolution of 10 m. The multispectral data, obtained from GF-6, were also resampled to a spatial resolution of 10 m. The dataset is annotated with 13 classes: "Surface Water," "Street," "Urban," "Industrial," "Mines," "Artificial Vegetation," "Arable Land," "Permanent Crops," "Pastures," "Forests," "Shrubs," "Open Spaces," and "Wetlands." Following the same settings [11], the Beijing data were cropped into 256 × 256 pixel patches with an overlap ratio of 0.5 for training. The Wuhan data were cropped into 256 × 256 pixel patches without overlap for testing. Consequently, the training and testing sets comprised 7140 and 850 pairs, respectively.

B. Experimental Settings

1) Implementation Details: In the proposed REF-Net, two ResNet-50 [51] pre-trained on ImageNet [52] are utilized as pseudo-Siamese feature extractors. Data augmentation includes random rotation and random horizontal flips. The model is optimized using the AdamW optimizer, with a weight decay set to 0.0005. The batch size is set to 8, and the initial learning rate is 0.0005. The model is trained for 70 000 iterations on the DFC20 dataset, 150 000 iterations on the WHU-OPT-SAR dataset, and 80 000 iterations on the C2Seg-BW dataset. A polynomial scheduler is employed to reduce the learning rate according to the formula: $\text{lr} = \text{base_lr} \times (1 - \text{iter}/\text{max_iter})^{0.9}$. The random seed is set to 1 for all experiments. All methods are implemented using PyTorch 2.0.1 on an Ubuntu 20.04 platform, utilizing an NVIDIA RTX A6000 GPU (48 GB).

2) Evaluation Metrics: Following representative studies [12], [13], [14], [15], [53], four widely used metrics are utilized to evaluate the performance of REF-Net: overall accuracy (OA), mean $F1$-score (mF1), kappa coefficient (Kappa), and mean intersection over union (mIoU). Here, mF1 and mIoU represent the average $F1$-score and IoU values across all categories. The calculation of these metrics is as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{8}$$

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{10}$$

$$\text{Kappa} = \frac{\text{OA} - p_e}{1 - p_e}$$

$$p_e = \frac{(\text{TN} + \text{FN}) \times (\text{TN} + \text{FP})}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} + \frac{(\text{TP} + \text{FP}) \times (\text{FN} + \text{TP})^2}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{11}$$

where TP, TN, FP, and FN are the true positives, true negatives, false positives, and false negatives.

3) Comparison Methods: To verify the superiority of the proposed REF-Net in multimodal RS image LULC classification, several representative methods were selected for comparison, including: MCANet [15], JoiTriNet [13] (including JoiTriNet-e and JoiTriNet-d), SAMFNet [14], FTransUnet [53], CEN [54], MultiSenseSeg [55], Sigma [56], and ASD [12]. Following [42], [57], and [58], these compared methods are trained based on the training settings in the original papers.

C. Experiments on DFC20 Dataset

1) Performance Comparison: In this section, comprehensive experiments are conducted to quantitatively compare REF-Net with other representative methods on the DFC20 dataset. Experimental results for these methods are summarized in Table I. Additionally, the complexity comparison with compared methods is reported in Table II, involving parameters, computation complexity (FLOPs), and speed (FPS).

As shown in Table I, columns 1–5, compared to these representative methods, REF-Net achieves substantial improvements across all metrics, with increases of 1.19%–4.83% in OA, 1.83%–7.75% in mF1, 1.44%–5.92% in Kappa, and 2.55%–9.94% in mIoU. These results highlight the effectiveness of REF-Net in multimodal RS image land cover classification. Table I, columns 6–13, reports the IoU values for each category. Notably, the categories "Shrublands," "Wetlands," and "Barren" are particularly challenging to classify. REF-Net significantly improves the IoU by 4.00%–15.09% for "Shrublands," 2.70%–10.85% for "Wetlands," and 3.30%–17.73% for "Barren." These results show that the proposed REF-Net achieves the highest IoU performance across all categories in the DFC20 dataset, demonstrating its superior ability to distinguish various land cover types. Apart from these competitive classification results, Table II indicates that the proposed REF-Net achieves a trade-off between classification performance and computational complexity.

2) Visualization of the Proposed Method: To intuitively evaluate the classification performance, several representative scenarios are selected. Visual comparisons between REF-Net

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TABLE I

COMPARISON WITH MULTIMODAL FUSION METHODS ON DFC20 DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	OA (%)	mF1 (%)	Kappa (%)	mIoU (%)	Category IoU (%)							
					Forests	Shrub-lands	Grass-lands	Wet-lands	Crop-lands	Urban/build_up	Barren	Water
MCANet [15]	85.64	76.96	82.44	64.69	84.42	46.73	60.27	50.86	69.24	70.94	36.79	98.31
JoiTriNet-e [13]	86.63	78.92	83.66	66.95	85.58	51.08	61.57	53.51	71.12	71.86	42.44	98.43
JoiTriNet-d [13]	86.78	78.99	83.83	67.07	85.52	51.14	62.18	53.86	71.70	71.93	41.85	98.41
SAMFNet [14]	88.58	81.31	86.06	70.21	88.60	54.68	67.42	56.32	74.37	77.14	44.33	98.85
FTransUnet [53]	87.59	80.03	84.85	68.50	87.53	51.57	65.10	54.36	72.25	75.14	43.45	98.61
CEN [54]	88.10	81.23	85.46	69.89	87.95	54.54	65.65	56.68	73.24	74.31	48.08	98.69
MultiSenseSeg [55]	88.70	81.99	86.20	70.93	88.55	55.62	67.28	57.71	74.47	76.06	48.89	98.84
Sigma [56]	89.20	<u>82.88</u>	86.81	<u>72.08</u>	89.23	<u>57.82</u>	68.51	<u>59.01</u>	75.40	76.57	<u>51.22</u>	98.90
ASD [12]	<u>89.28</u>	82.33	<u>86.92</u>	71.57	<u>89.79</u>	56.87	<u>69.00</u>	57.55	<u>75.79</u>	<u>78.31</u>	46.32	<u>98.91</u>
REF-Net (proposed)	**90.47**	**84.71**	**88.36**	**74.63**	**90.76**	**61.82**	**71.57**	**61.71**	**77.80**	**79.75**	**54.52**	**99.08**

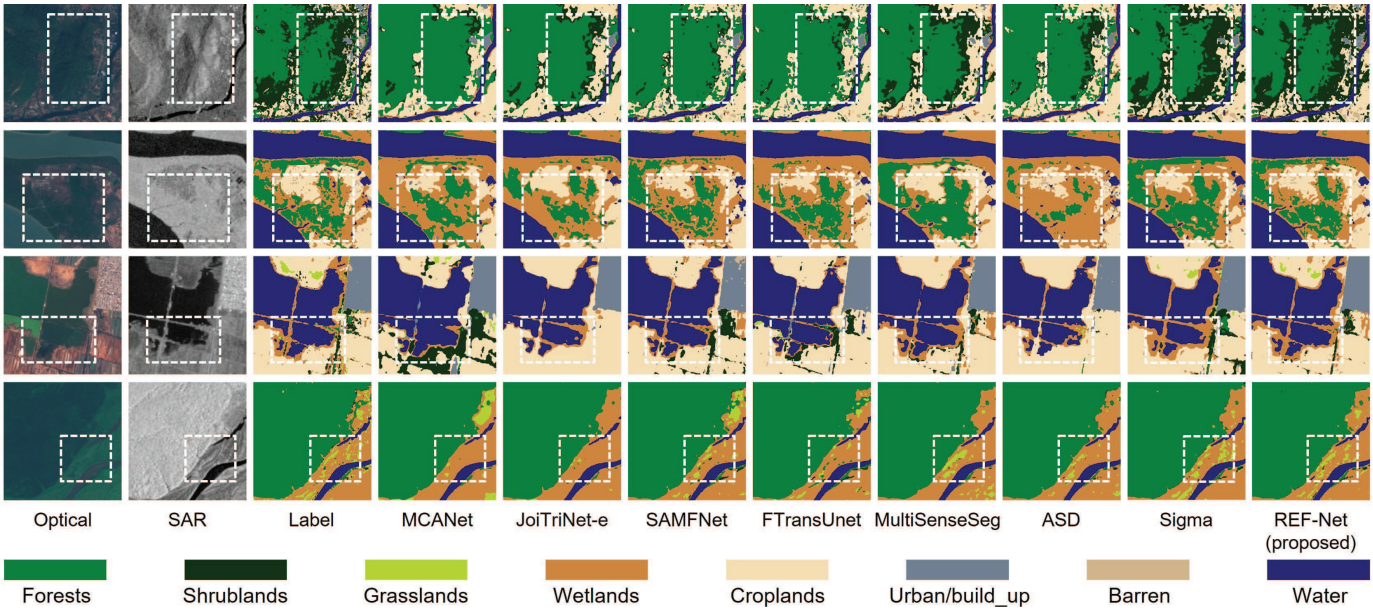


Fig. 6. Visualization of the proposed REF-Net and compared representative methods on DFC20 dataset.

TABLE II

COMPLEXITY COMPARISON WITH MULTIMODAL FUSION METHODS WHEN OPTICAL IMAGE SIZE IS $3 \times 256 \times 256$ AND SAR IMAGE SIZE IS $2 \times 256 \times 256$

Method	OA (%)	Params (M)	FLOPs (G)	Speed (FPS)
MCANet [15]	85.64	213.94	226.82	27.21
JoiTriNet-e [13]	86.63	104.10	107.29	35.11
JoiTriNet-d [13]	86.78	91.49	95.67	36.74
SAMFNet [14]	88.58	19.60	28.78	101.22
FTransUnet [53]	87.59	179.42	56.14	25.07
CEN [54]	88.10	118.14	131.34	39.19
MultiSenseSeg [55]	88.70	57.73	66.09	36.76
Sigma [56]	89.20	57.79	24.16	28.13
ASD [12]	89.28	87.01	22.04	36.42
REF-Net (proposed)	90.47	54.86	48.43	27.41

and other representative methods on the DFC20 dataset are shown in Fig. 6.

In Fig. 6, rows 1–3 depict optical images containing spectral or textural information, and SAR images are less discriminative. Most other methods struggle to accurately classify these land cover objects. Specifically, the edges of "Shrublands" in row 1 are misclassified as "Forests," "Wetlands" in row 2 are confused with "Forests," and the boundaries of "Wetlands" in row 3 are unclear, likely due to confusion with "Croplands." Fig. 6, row 4, shows an optical image with blurred texture and a SAR image with relatively clear boundaries. These compared methods fail to accurately distinguish "Grasslands" from "Wetlands." In contrast, REF-Net produces promising results, intuitively demonstrating its competitive performance on multimodal RS image LULC classification.

3) Error Bar Visualization: To evaluate the randomness, all the compared methods were repeated five times [59] with different random seeds on the DFC20 dataset. The error bar visualization, which includes the mean and standard deviation values of OA, is shown in Fig. 7. The proposed REF-Net achieves high mean values and relatively small error fluctuations, demonstrating its robustness and consistency.

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TABLE III
EFFECTIVENESS OF THE PROPOSED MDF AND MSIE DECODERS

MDF decoder	MSIE decoders		Params (M)		FLOPs (G)		Speed		OA (%)	mF1 (%)	Kappa (%)	mIoU (%)
	Auxiliary classification	Modality-difference feature	Train	Infer	Train	Infer	Train Hours	Infer FPS				
✗	✗	✗	51.38	51.38	28.81	28.81	4.90	52.14	89.74	83.57	87.47	73.06
✓	✗	✗	54.86	54.86	48.43	48.43	12.17	27.41	90.18	84.26	88.02	74.00
✓	✓	✗	57.52	54.86	61.36	48.43	12.73	27.41	90.25	84.36	88.10	74.15
✓	✓	✓	60.48	54.86	74.27	48.43	13.42	27.41	90.47	84.71	88.36	74.63

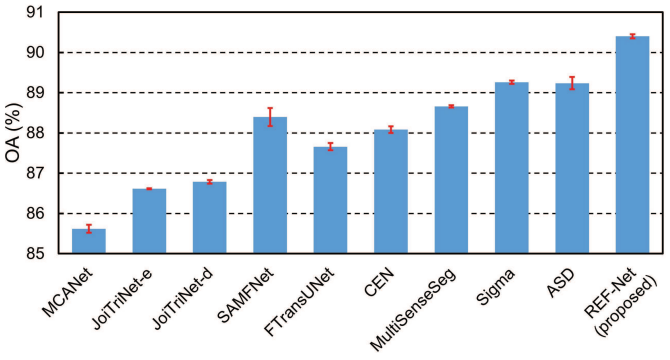


Fig. 7. Error bar visualization of compared methods on DFC20 dataset.

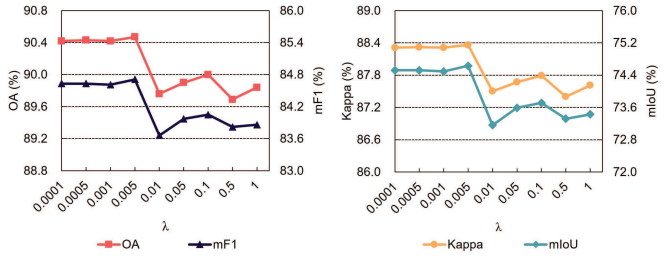


Fig. 8. Verification of hyperparameter λ. The highest performance is achieved when λ is set to 0.005.

4) Hyperparameter Analysis: The sole hyperparameter in the proposed REF-Net is λ, which controls the contribution of the MSIE decoder. To assess the impact of λ, hyperparameter analysis experiments are conducted using the DFC20 dataset. Specifically, λ is varied across a range from 0.0001 to 1.0. As illustrated in Fig. 8, OA, mF1, Kappa, and mIoU values exhibit promising performance when λ is below 0.005. However, performance declines when λ exceeds 0.005, with optimal results observed at $\lambda = 0.005$. Consequently, λ is set to 0.005 for all datasets.

5) Effectiveness of the Proposed Decoders: To assess the effectiveness of the proposed MDF and MSIE decoders, verification experiments are conducted. Table III, rows 2, 3, and 5, presents the results for the baseline, the MDF decoder, and the combined MDF and MSIE decoders, respectively.

a) Baseline analysis: The baseline model achieves an OA of 89.74%, mF1 of 83.57%, Kappa of 87.47%, and mIoU of 73.06%, outperforming the compared methods in Table I. The high performance of the baseline can be attributed to the dataset characteristics and the decoder design. As shown in Fig. 6, the DFC20 dataset provides detailed annotations, particularly along object boundaries, which favor methods with high-resolution feature maps. Among the compared methods, MCANet, JoiTriNet-e/d, CEN, MultiSenseSeg, and Sigma use a decoder feature map stride of 4, which results in insufficient detail for land cover information. SAMFNet and ASD use a stride of 2, while FTransUnet uses a stride of 1; however, these models fail to capture complex global contextual information. In contrast, the baseline model uses a decoder feature map stride of 2 with a VSS block at each stage, allowing it to capture complex global context and generate high-resolution feature maps, which contributes to its superior performance.

b) Decoder analysis: First, incorporating the MDF decoder on baseline (Table III, rows 2 and 3) significantly improves the performance, yielding increases of 0.94% in mIoU, 0.44% in OA, 0.69% in mF1, and 0.55% in Kappa. These improvements require an additional 3.48 M parameters, increasing FLOPs to 48.43 G, while remaining comparable to other methods in Table II. Therefore, the MDF decoder significantly enhances performance while maintaining complexity comparable to other methods. Then, after incorporating the MSIE decoder (Table III rows 3 and 5), additional improvements of 0.63% in mIoU, 0.29% in OA, 0.45% in mF1, and 0.34% in Kappa are observed, with an increase of 5.62 M parameters, 25.84 GFLOPs, and 1.25 h of training time. Meanwhile, the inference complexity remains at 48.43 GFLOPs. Therefore, the MSIE decoder achieves substantial classification improvements with an acceptable training cost for model development. These improvements confirm the effectiveness of MDF and MSIE decoders, thus retaining and enhancing modality-specific information beneficial to classification. Additionally, Fig. 9 shows that the training losses of the MDF decoder and MSIE decoder decrease consistently, demonstrating a stable training process.

c) Modality-difference feature in the MSIE decoder: In the MSIE decoder, modality-difference features are introduced into multimodal features to perform auxiliary classification tasks. Thus, the MSIE decoder contains two main operations: auxiliary classification and modality-difference feature introduction. To further investigate the impact of these two operations in the MSIE decoder, verification experiments are conducted. Table III, rows 4 and 5, presents the results for auxiliary classification and modality-difference feature introduction.

First, performing auxiliary classification (Table III rows 3 and 4) yields only slight improvements: 0.15%

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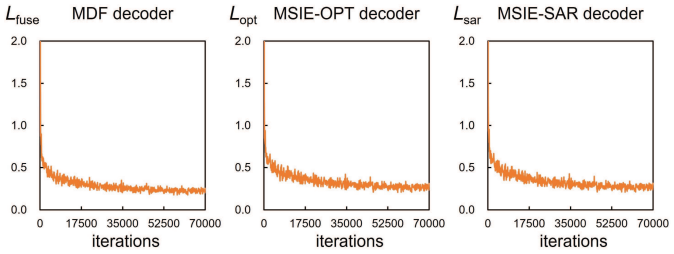


Fig. 9. Loss curve of the proposed MDF and MSIE decoders on DFC20 dataset.

TABLE IV
PROJECTION DIMENSION ANALYSIS IN THE PROPOSED REF-NET DURING TRAINING

Dim	Params (M)	FLOPs (G)	OA (%)	mF1 (%)	Kappa (%)	mIoU (%)
64	50.65	26.98	89.85	83.67	87.61	73.23
128	60.48	74.27	90.47	84.71	88.36	74.63
256	98.71	262.11	90.68	85.01	88.63	75.06

in mIoU, 0.10% in mF1, 0.07% in OA, and 0.08% in Kappa. Then, after introducing modality-difference features to auxiliary classification (Table III rows 4 and 5), these metrics are further improved by 0.48% in mIoU, 0.35% in mF1, 0.22% in OA, and 0.26% in Kappa. The improvements of these metrics, particularly in mIoU and mF1, which are sensitive to difficult objects, indicate that introducing modality-different features into auxiliary classification, i.e., modality-specific information enhancement, is crucial for the MSIE decoder.

d) Real-time performance: As shown in Table II, the computational cost of REF-Net is comparable to other advanced methods such as FTransUnet and MultiSenseSeg, but remains higher than Sigma and ASD, indicating potential for further efficiency optimization. Real-time applications typically require a processing speed of at least 30 FPS [60], while REF-Net achieves 27.41 FPS, which is slightly below the real-time threshold and therefore requires improvement.

e) Feasibility of deployment: REF-Net employs only an MDF decoder during inference and deployment, maintaining a relatively compact structure. Its modular design provides flexibility for further optimization via network pruning [61], lightweight backbone replacement [62], or quantization [63]. These techniques can significantly enhance real-time performance and enable deployment on mobile or resource-constrained platforms.

6) Projection Dimension Analysis: In the proposed REF-Net, optical and SAR features are projected into the same dimension, which is empirically set to 128. In this section, comparative experiments are conducted to analyze the impact of the projection dimension. The projection dimension is set to 64, 128, and 256 to evaluate classification performance and computation complexity, as reported in Table IV.

According to Table IV, row 2, when the projection dimension is set to 64, REF-Net achieves 89.85% in OA, 83.67% in mF1, 87.61% in Kappa, and 73.23% in mIoU, with 50.65 M parameters and 26.98 GFLOPs. From Table IV, rows 2 and 3, it can be observed that when the dimension is set to 128, REF-Net shows significant improvements, with increases of 1.40% in mIoU, 1.04% in mF1, 0.75% in Kappa, and 0.62% in OA. Meanwhile, the parameters and FLOPs are 60.48 M and 74.27 G, which are acceptable. However, when the projection dimension is set to 256, the improvements are slight, with increases of 0.21% in OA, 0.30% in mF1, 0.27% in Kappa, and 0.43% in mIoU. Meanwhile, the parameters and FLOPs rise to 98.71 M and 262.11 G, respectively, representing approximately 1.63 and 3.53 times increases in storage and computational complexity. Therefore, to balance performance and computational cost, the projection dimension is set to 128 for all experiments.

TABLE V
EFFECTIVENESS OF THE VSS BLOCK UTILIZED IN THE PROPOSED MDF-VSS BLOCK AND MDF DECODER. THE BEST RESULTS ARE MARKED IN BOLD

Module	OA (%)	mF1 (%)	Kappa (%)	mIoU (%)
Res block	86.95	78.95	84.03	67.13
ViT block	89.14	82.43	86.74	71.58
VSS block	**90.18**	**84.26**	**88.02**	**74.00**

7) Effectiveness of the VSS Block: VSS blocks are utilized to develop the MDF-VSS block and MDF decoder, aiming to capture the global spatial context of complex land objects. To assess the effectiveness of VSS blocks within the MDF decoder, they are replaced with widely used residual blocks and ViT blocks, representing CNN and Transformer architectures, respectively. Comparative experiments are conducted, and the results are presented in Table V.

As shown in Table V, rows 2 and 3, the ViT block outperforms the residual block by 2.19% in OA, 3.48% in mF1, 2.71% in Kappa, and 4.45% in mIoU, respectively. Also, as shown in Table V, rows 2 and 4, the VSS block surpasses the residual block by 3.23% in OA, 5.31% in mF1, 3.99 % in Kappa, and 6.87% in mIoU, respectively. Compared to the Res block, both ViT and VSS blocks are more effective at capturing global spatial context, leading to significant improvements. This indicates that capturing global contextual information is essential in this architecture.

Then, as shown in Table V, rows 3 and 4, the VSS block outperforms the ViT block by 1.04% in OA, 1.83% in mF1, 1.28% in Kappa, and 2.42% in mIoU, respectively. This result demonstrates that the VSS block is much effective in REF-Net and achieves superior performance. ViT block divides features into small patches, which may lose spatial information [41]. In contrast, the VSS block handles entire features using a scan and merge mechanism. This approach captures relationships between pixels and elaborately extracts global contextual information from complex land objects, thereby boosting performance.

D. Experiments on WHU-OPT-SAR Dataset

1) Performance Comparison: In this section, extensive experiments are conducted to quantitatively compare the proposed REF-Net with other representative methods on the

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TABLE VI

COMPARISON WITH MULTIMODAL FUSION METHODS ON WHU-OPT-SAR DATASET. THE BEST AND SECOND-BEST ACCURACIES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	OA (%)	mF1 (%)	Kappa (%)	mIoU (%)	Category IoU (%)						
					farmland	city	village	water	forest	road	others
MCANet [15]	81.40	61.05	71.23	47.45	64.47	47.76	43.89	55.92	81.94	29.50	8.64
JoiTriNet-e [13]	82.41	65.05	73.06	51.05	66.74	52.78	45.50	59.95	82.11	34.67	15.64
JoiTriNet-d [13]	82.07	64.69	72.62	50.66	65.75	53.29	45.45	59.23	82.10	33.08	15.74
SAMFNet [14]	83.15	66.32	74.18	52.45	67.67	54.89	47.70	61.28	82.87	36.59	16.16
FTransUnet [53]	83.23	66.80	74.35	52.88	67.98	55.23	48.23	61.07	82.96	37.50	17.20
CEN [54]	82.69	65.76	73.47	51.77	66.85	54.65	46.61	60.10	82.53	35.00	16.68
MultiSenseSeg [55]	83.14	66.69	74.25	52.69	68.15	52.19	48.35	61.72	82.67	38.10	17.61
Sigma [56]	83.33	67.45	74.53	53.44	67.67	54.67	47.62	62.27	83.15	**39.98**	18.74
ASD [12]	83.27	66.76	74.33	52.79	68.10	54.40	47.19	61.79	82.85	37.30	17.89
REF-Net (proposed)	**83.92**	**68.30**	**75.46**	**54.38**	**68.99**	**55.87**	**49.35**	**63.19**	**83.65**	39.58	**20.04**

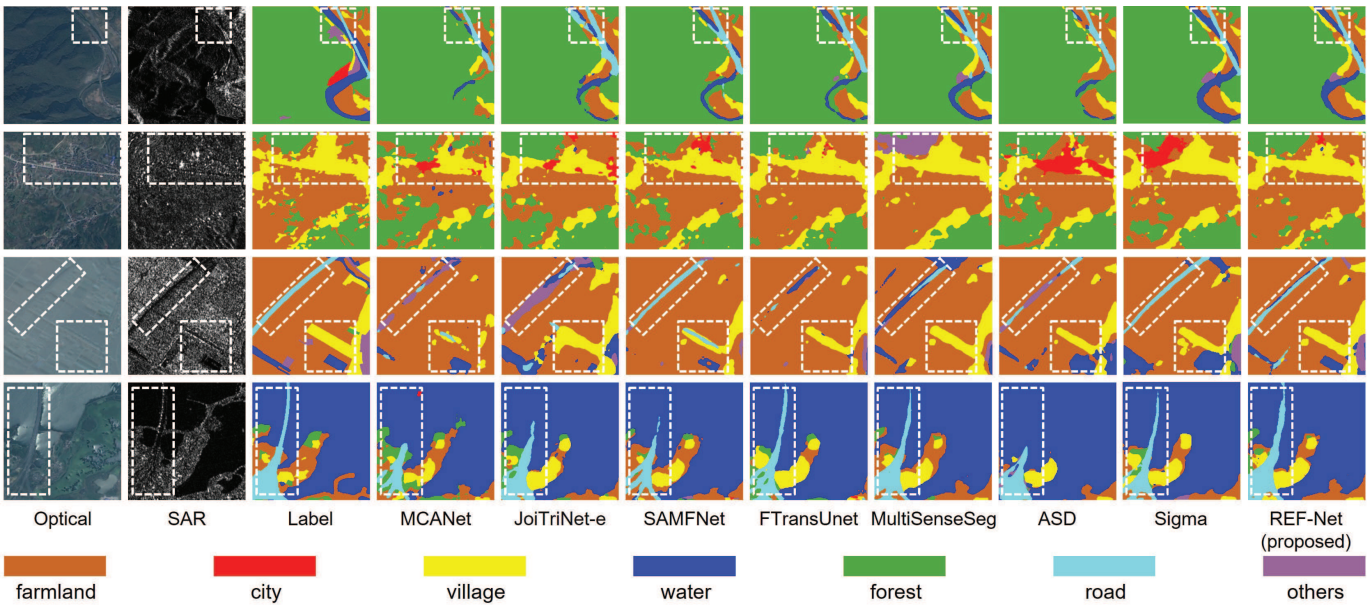


Fig. 10. Visualization of the proposed REF-Net and compared methods on the WHU-OPT-SAR dataset.

WHU-OPT-SAR dataset. Experimental results are reported in Table VI.

As shown in Table VI, columns 1–5, compared to other representative methods, REF-Net shows significant improvements across all metrics, with an OA increase of 0.59%–2.52%, mF1 by 0.85%–7.25%, Kappa by 0.93%–4.23%, and mIoU by 0.94%–6.93%. These results demonstrate the effectiveness of REF-Net in multimodal RS image land use classification. Table VI, columns 6–12, reports the IoU values for each category. Notably, the categories "city," "village," and "water" are particularly challenging to classify. The proposed REF-Net significantly improved the IoU by 0.64%–8.11% for "city," 1.00%–5.46% for "village," and 0.92%–7.27% for "water," respectively. These results show that REF-Net achieves the best or second-best IoU performance across all categories, demonstrating its superior ability to distinguish various land use types.

2) Visualization of the Proposed Method: To visually assess classification performance, several representative scenarios are selected. The visual comparisons between the proposed REF-Net and other representative methods are shown in Fig. 10.

In Fig. 10, rows 1 and 2 show optical images with spectral and texture information, and SAR images cannot describe land objects clearly. These compared methods struggle to classify these land use objects. Specifically, "water" in row 1 is incoherent and confused with "road," and "village" in row 2 is classified as "city." Fig. 10, rows 3 and 4, presents optical images with blurred textures and SAR images with clear backscatter information, effectively revealing the boundaries of "road" and "village." However, these compared methods fail to segment "road" or "village" with fine details. In contrast, REF-Net provides promising results, clearly demonstrating its competitive performance on multimodal RS image LULC classification.

3) Feature Comparison and Analysis: To intuitively evaluate the mechanism of REF-Net, CAM visualizations are performed on the WHU-OPT-SAR dataset. For comparison, the representative method ASD [12] is also evaluated in this

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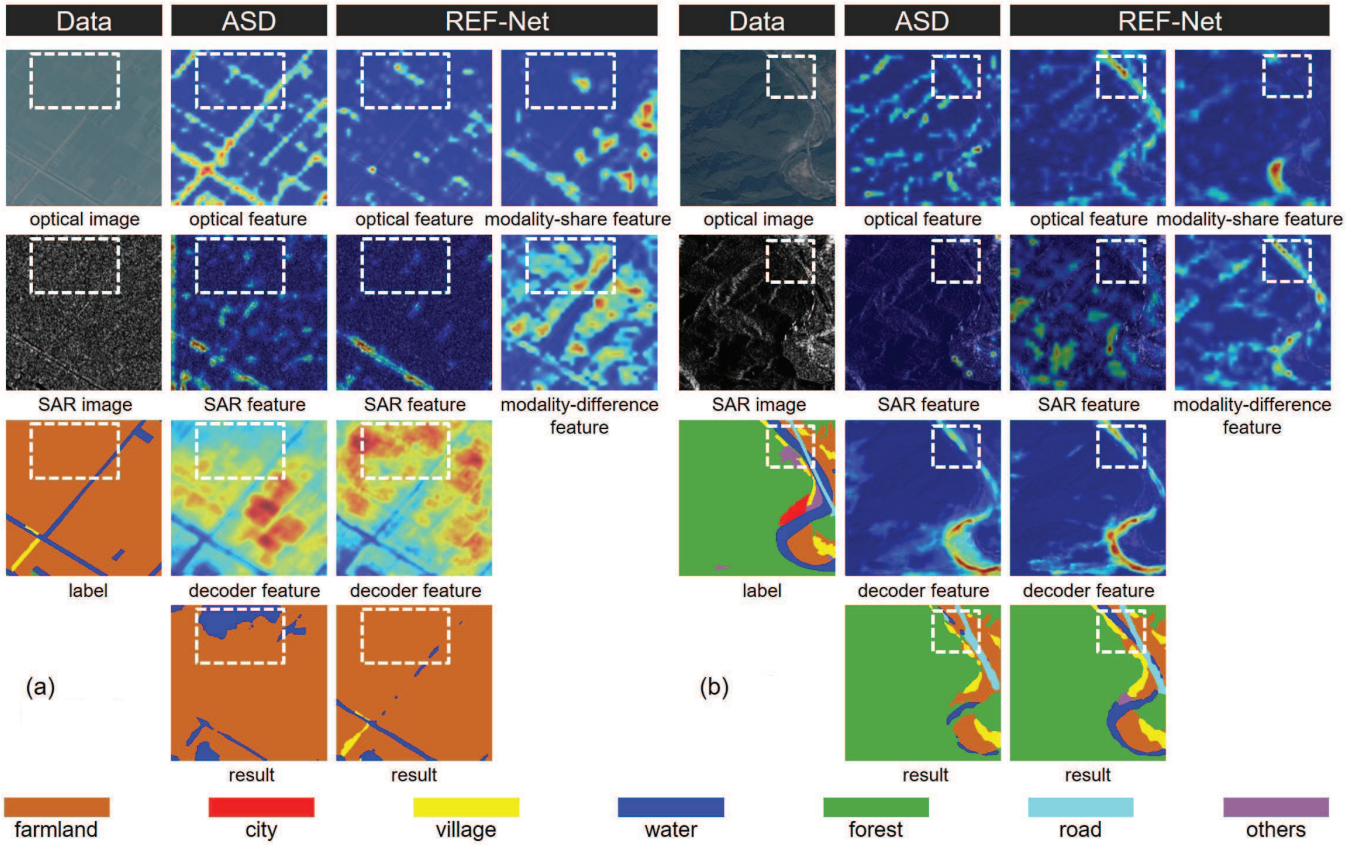


Fig. 11. CAM visualization of representative natural scenarios of (a) "farmland" and (b) "water" for representative method ASD and the proposed REF-Net.

experiment. These visualizations include multimodal features, decoder features, as well as modality-share and modality-difference features.

First, representative natural scenarios are selected for CAM visualization of features, as shown in Fig. 11. Fig. 11(a) illustrates the "farmland" category. As indicated by the white box, the SAR image is less discriminative, while the optical image clearly shows farmland texture. In the CAM visualization of optical and SAR features, both ASD and REF-Net exhibit minimal responses in the SAR features but show strong responses in the optical image. Thus, optical features contain crucial modality-specific information for classifying "farmland." However, ASD shows weak responses in the decoder, causing confusion with the "water" class. This suggests that ASD loses optical-specific information essential for accurate classification. In REF-Net, distinct responses in the optical features are highlighted in the modality-difference feature, which indicates that the critical modality-specific information is emphasized. Then, the common response area in both optical and SAR features is captured by the modality-share features, indicating shared information perceivable by both modalities. The responses in shared and difference features are also reflected in the decoder features. These results verify that REF-Net effectively alleviates the loss of optical-specific information critical for classification, thereby increasing classification performance.

Fig. 11(b) illustrates the "water" category. As indicated by the white box, the SAR image is difficult to distinguish, whereas the optical image provides relatively clear textures, which are crucial for classification. In the CAM visualization of ASD and REF-Net, SAR features show weak responses to "water," while optical features respond clearly, suggesting that optical-specific information aids classification. The decoder feature of ASD shows incomplete responses for "water" at the bottom, and the classification result is misclassified as "road" and "farmland," indicating that some optical-specific information is lost in this case. In REF-Net, the modality-difference feature accurately outlines the "water" area, indicating that beneficial modality-specific information is effectively captured. Then, the modality-share feature shows minimal activation due to weak responses in the SAR feature. These responses are also shown in the decoder features, confirming that REF-Net alleviates the loss of optical-specific information essential for classification, thereby increasing the classification performance.

Then, representative artificial scenarios are selected for CAM visualization of features, as shown in Fig. 12. Fig. 12(a) illustrates the "road" category. As indicated by the white box, the top of the optical image presents similar spectral properties and smooth textures for both "road" and "water," while the bottom clearly distinguishes the "road" and "village." The top of the SAR image captures backscatter information, accurately

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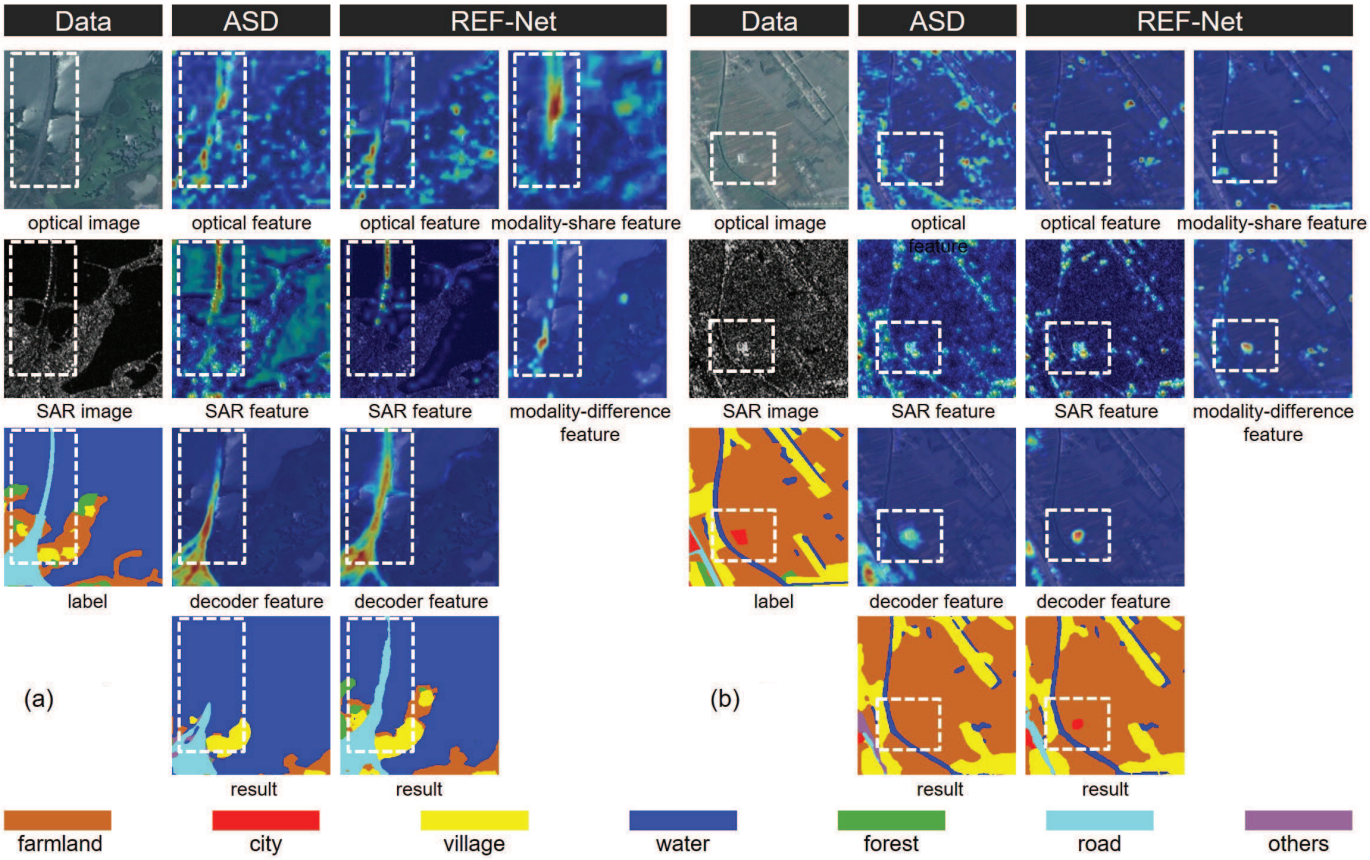


Fig. 12. CAM visualization of representative artificial scenarios of (a) "road" and (b) "city" for representative method ASD and the proposed REF-Net.

reflecting the road structure, while the bottom is less distinguishable. CAM visualization for ASD and REF-Net shows strong responses from the optical features at the bottom and the SAR features at the top, indicating the optical- and SAR-specific information. It is evident that ASD loses SAR-specific information in the decoder feature, leading to misclassification in the final result. In REF-Net, the responses in the modality-share feature align with the area where both optical and SAR features respond, indicating perceivable by both modalities. Then, the responses of the modality-difference feature, containing both optical- and SAR-specific information, accurately delineate the "road," emphasizing critical specific information. These responses are preserved in the decoder feature and show complement characteristics. These results confirm that REF-Net effectively alleviates the loss of modality-specific information and achieves satisfying classification results.

Fig. 12(b) illustrates the "city" category. The white box shows that the optical image presents similar spectral or texture information for both "village" and "city," while the SAR image clearly delineates the structure of "city." In the CAM visualization, both ASD and REF-Net exhibit weak responses to optical features but strong responses to SAR features. This indicates that the SAR features contain SAR-specific information crucial for classifying the "city" category. In ASD, the decoder exhibits weak responses and misclassifies the area as "farmland." It can be speculated that ASD loses some SAR-specific information crucial for classification. In REF-Net, the modality-difference feature captures "city" with strong responses, effectively emphasizing the beneficial modality-specific information. Then, the modality-share feature shows minimal activation due to the limited responses in the optical feature. These responses are also reflected in the decoder features, confirming that REF-Net alleviates the loss of modality-specific information, leading to satisfying classification results.

In summary, the visualization experiments consistently demonstrate that REF-Net effectively decomposes modality-share and modality-difference features while controlling the contribution of beneficial shared and specific information. This helps alleviate the loss of modality-specific information, resulting in improved classification results.

4) Feature Numerical Analysis: In this section, the comparability of multimodal feature value magnitudes has been validated. Fig. 13 shows the distribution of optical and SAR feature values processed by the MDF decoder on the WHU-OPT-SAR dataset. Here, the x-axis represents the feature values, and the y-axis represents the frequency of each value. The offsets between the optical and SAR feature distributions reflect differences in modality-specific information. The differences in feature values between the distributions have comparable magnitudes, making them effectively measurable through the subtraction operation.

5) Conditions of Modality Decomposition: In the MDF-VSS block, modality decomposition requires that at least

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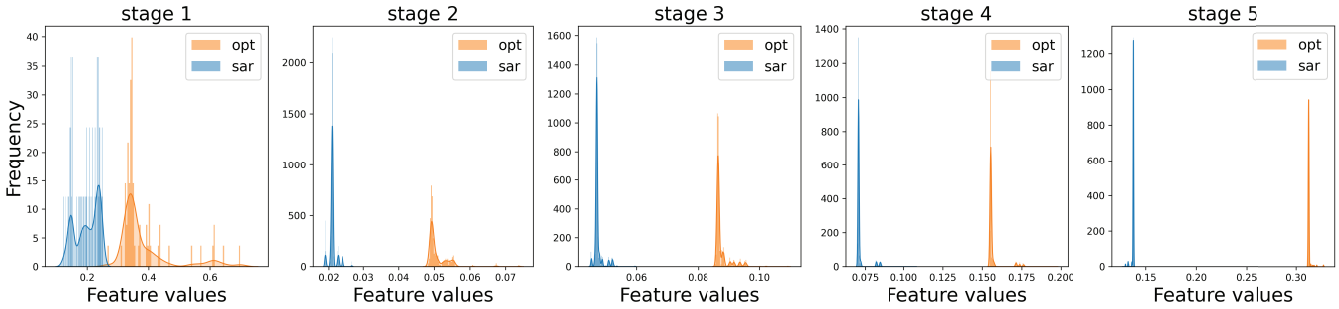


Fig. 13. Distribution of optical and SAR feature values processed by the MDF decoder on the WHU-OPT-SAR dataset.

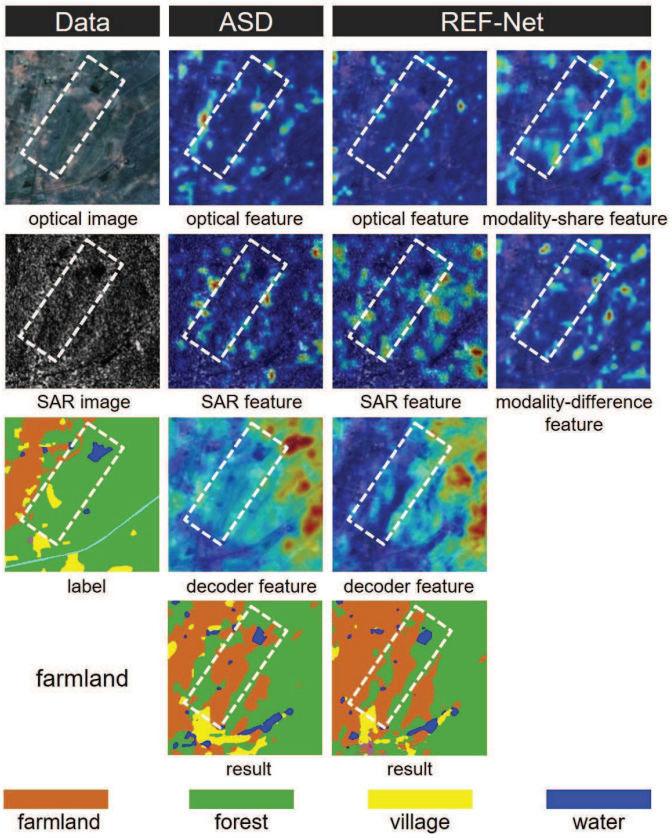


Fig. 14. CAM visualization of "forest" scenario, where both optical and SAR data fail to provide distinctive information.

one of the modalities provides critical specific information for classification. If neither modality provides discriminative specific information, this block may struggle to effectively decompose distinct modality-share and modality-difference features.

Fig. 14 presents a case from the WHU-OPT-SAR dataset, where optical and SAR images exhibit complex distributions for the "forest" class. In the CAM visualization, ASD and REF-Net show relatively weak activations for the "forest" class in both optical and SAR features. Under such circumstances, the decomposed modality-share and modality-difference features in REF-Net exhibit weak responses, leading to weak activations in the decoder feature and incorrect classification results.

TABLE VII

COMPARISON WITH MULTIMODAL FUSION METHODS ON C2SEG-BW DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	OA (%)	mF1 (%)	Kappa (%)	mIoU (%)
MCANet [15]	42.21	19.23	33.04	13.30
JoiTriNet-e [13]	42.40	<u>22.00</u>	33.53	<u>14.76</u>
JoiTriNet-d [13]	<u>42.69</u>	21.47	<u>34.43</u>	14.75
SAMFNet [14]	41.28	20.11	32.33	13.60
FTransUnet [53]	42.61	21.44	33.58	14.34
CEN [54]	40.17	18.36	31.55	12.69
MultiSenseSeg [55]	41.90	20.15	32.53	13.65
Sigma [56]	35.43	18.32	27.11	12.26
ASD [12]	40.94	20.55	32.07	13.83
REF-Net (proposed)	**42.82**	**22.86**	**34.53**	**15.50**

E. Experiments on C2Seg-BW Dataset

1) Performance Comparison: This section presents comprehensive experiments to further evaluate the REF-Net and other methods on the C2Seg-BW dataset. The experimental results of these methods are summarized in Tables VII and VIII.

As shown in Tables VII and VIII, this dataset presents a challenge to all methods, resulting in suboptimal performance across the board. Compared to other methods, REF-Net shows improvements across all metrics, with increases of 0.86%–4.54% in mF1, 0.74%–3.24% in mIoU, 0.13%–7.39% in OA, and 0.10%–7.42% in Kappa. Then, REF-Net achieves the best or second-best category IoU results on five classes. Among the compared methods, the Sigma method shows low performance. A possible explanation for this is that Sigma employs VSS blocks in both the encoder and decoder, which might be too dependent on a specific scan order. This reliance could make it challenging to handle this dataset with distribution shifts, leading to degraded performance.

2) Visualization of the Proposed Method: To visually assess the classification performance, several scenarios are selected for comparison. Visual comparisons between the proposed REF-Net and other methods are shown in Fig. 15.

In Fig. 15, rows 1 and 2 show that optical images fail to clearly describe "Urban" and "Industrial," while the corresponding SAR images contain backscatter information that helps distinguish them. The compared methods have difficulty accurately classifying "Urban" and "Industrial." Rows 3 and 4

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TABLE VIII

CATEGORY IOU COMPARISON WITH MULTIMODAL FUSION METHODS ON C2SEG-BW DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Category IoU (%)												
	Surf. Water	Street	Urban	Ind.	Mines	Art. Veg.	Arable Land	Crops	Past.	Forests	Shrubs	Open Sp.	Wet/ lands
MCANet [15]	72.79	23.92	20.43	13.69	**1.43**	5.36	5.33	-	-	29.89	-	-	-
JoiTriNet-e [13]	69.47	20.82	23.38	**23.97**	0.60	5.93	12.40	1.30	-	26.68	7.21	0.19	-
JoiTriNet-d [13]	69.17	18.78	14.05	20.28	0.70	7.09	**20.33**	0.14	-	**40.93**	0.23	0.01	**0.03**
SAMFNet [14]	66.24	19.80	21.93	16.10	0.56	6.95	0.02	-	-	34.26	**10.20**	**0.68**	-
FTransUnet [53]	60.29	22.73	**28.01**	16.67	0.31	**7.68**	13.00	0.54	-	36.09	1.02	0.02	**0.03**
CEN [54]	71.16	13.10	11.96	18.56	0.51	4.33	5.09	-	-	36.38	3.76	0.06	-
MultiSenseSeg [55]	**72.99**	21.20	18.57	11.12	0.60	6.67	12.68	-	0.58	26.93	5.87	0.17	-
Sigma [56]	65.99	22.99	21.51	16.39	0.76	5.86	1.45	0.04	**0.82**	19.32	3.92	0.34	-
ASD [12]	66.99	21.99	21.15	20.59	0.52	6.69	2.95	**2.21**	0.56	31.19	4.85	0.11	0.02
REF-Net (proposed)	68.38	**26.69**	22.37	17.73	0.80	6.64	14.66	0.51	0.15	37.23	5.86	0.46	0.01

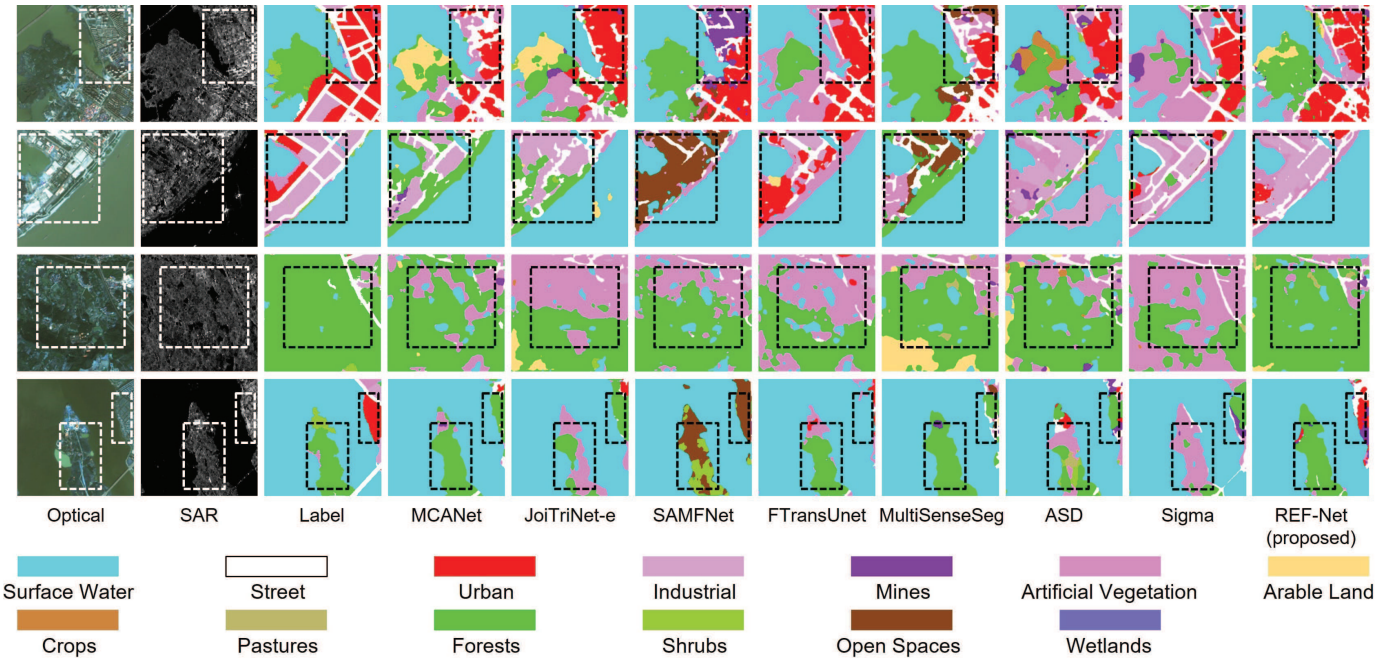


Fig. 15. Visualization of the proposed REF-Net and compared methods on the C2Seg-BW dataset.

in Fig. 15 showcase optical images with spectral information for "Forest" and "Artificial Vegetation," whereas SAR images struggle to classify them. In this case, the compared methods fail to segment "Forest" in some challenging scenes. In contrast, REF-Net produces better visualization results, especially in boxes.

F. Limitations of the Proposed Method

This study proposes a novel REF-Net to retain and enhance modality-specific information beneficial for classification. REF-Net has limitations mainly in two aspects.

The proposed MDF block requires that at least one modality provides critical modality-specific information; otherwise, it may struggle to effectively decompose modality-share and modality-specific features.

REF-Net shows limited real-time performance and relatively high computational complexity. However, its compact and modular architecture provides flexibility for further optimization and mobile deployment.

V. CONCLUSION

In this study, a REF-Net has been designed for multimodal RS image LULC classification using optical and SAR images to retain and enhance modality-specific information beneficial to classification. REF-Net is built upon a designed MDF-VSS block and consists of an MDF decoder and an MSIE decoder. The MDF-VSS block emphasizes beneficial modality-specific information and perceives shared land cover information through modality-difference and modality-share features, which are then fused to obtain discriminative fused features. Based on the MDF-VSS block, the MDF decoder adaptively controls the contribution of beneficial multiscale modality-share and modality-specific information, thereby retaining beneficial multiscale modality-specific information. The MSIE decoder

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performs modality-difference feature-guided auxiliary classification tasks to enhance optical- and SAR-specific information that is expert in classification. Therefore, the proposed REF-Net enhances the representation ability of fused features and improves the performance of multimodal RS image LULC classification. The quantitative experimental results evaluated on three datasets demonstrate the effectiveness and competitiveness of the proposed REF-Net. The CAM visualization highlights that the proposed REF-Net effectively alleviates the loss of modality-specific information.

In the future, we aim to extend this study to incorporate more modalities to further enhance comprehensive interpretation, such as a digital surface model with height information and hyperspectral imagery with rich spectral data. Then, we will further consider integrating the designed MDF-VSS block into multimodal RS foundation models, enhancing their ability to manage shared and specific information.

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